Problem 4

What is the gravitational potential of a Newtonian gravitational field of a body with mass $m = 1\,\mathrm{kg}$ at a distance $r = 6.67\,\mathrm{m}$?

Solution

The gravitational potential φ at a distance r from a mass m is given by the formula:

$$\varphi = -\frac{Gm}{r}$$

Where: - $G = 6.67 \times 10^{-11}\,\mathrm{N \cdot m^2/kg^2}$ is the gravitational constant, - $m = 1\,\mathrm{kg}$ is the mass of the object, - $r = 6.67\,\mathrm{m}$ is the distance from the mass.

Substitute the values into the formula:

$$\varphi = -\frac{6.67 \times 10^{-11}}{6.67}$$

Calculate:

$$\varphi = -10^{-11}\,\mathrm{J/kg}$$

Conclusion: The gravitational potential at a distance of $6.67\,\mathrm{m}$ from a $1\,\mathrm{kg}$ mass is $-10^{-11}\,\mathrm{J/kg}$.